

Solution: We will first reduce the given matrix to rref form.

$$\left[\begin{array}{cccccc} \boxed{1} & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & \boxed{2} & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & \boxed{1} & 7 \end{array} \right] \quad \text{The boxed posns. are pivots.}$$

1) Row 1 = Row 1 + 2x Row 3

$$\left[\begin{array}{cccccc} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array} \right]$$

2) Row 2 = Row 2 + Row 3

$$\left[\begin{array}{cccccc} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & \boxed{2} & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array} \right]$$

3) Row 1 = Row 1 - Row 2

$$\left[\begin{array}{cccccc} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array} \right]$$

4) Row 2 = Row 2 x $\frac{1}{2}$

$$\left[\begin{array}{cccccc} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array} \right]$$

The corresponding equations are:

$$x_1 + 6x_2 + 3x_4 = 0$$

$$x_3 - 4x_4 = 5$$

$$x_5 = 7$$

Basic variables: x_1, x_3, x_5 Free variables: x_2, x_4

General solution:

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases}$$

Numerical Note:

In general, the forward phase of row reduction takes much longer than the backward phase. An algorithm for solving a system is usually measured in flops (or floating point operations). A flop is one arithmetic operation (+, -, *, /) on two real floating point numbers. For an $n \times (n+1)$ matrix, the reduction to echelon form can take $2n^3/3 + n^2/2 - 7n/6$ flops (which is approximately $2n^3/3$ flops when n is moderately large - say, $n/50$). In contrast, further reduction to reduced echelon form needs almost n^2 flops. For further reading, read Matrix Computation by Golub and Van Loan.